

HIGH-PROBABILITY MATHEMATICS MOCK TEST

Set 2 • Comprehensive 33-Question Boards Dominator

Time: 3 Hours | Max Marks: 80 | Format: Based on Series EF1GH/5, PQ1RS/1 & Rationalized 2024 Pattern

SECTION A: MCQS (1 MARK EACH)

1. Matrix with 36 elements, the number of possible orders it can have is:

- (A) 13 (B) 3
(C) 5 (D) 9

2. A is a square matrix of order 3 and $|A| = 5$, then $|\text{adj } A|$ is:

- (A) 5 (B) 125
(C) 25 (D) -5

3. Principal value of $\cot^{-1}(-1/\sqrt{3})$ or $\tan^{-1}(-\sqrt{3})$ is:

- (A) $\pi/3$ (B) $-\pi/3$
(C) $2\pi/3$ (D) $2\pi/3$

4. Sum of order and degree of $d/dx [(dy/dx)^3] = 0$ is:

- (A) 2 (B) 3
(C) 5 (D) 0

5. If $|a + b| = |a - b|$, then vectors a and b are:

- (A) Orthogonal (B) Parallel
(C) Unit vectors (D) Collinear

6. Integrating factor of $(1-x^2)dy/dx + xy = ax$ is:

- (A) $\sqrt{1-x^2}$ (B) $1/\sqrt{1-x^2}$
(C) $1-x^2$ (D) $1/(1-x^2)$

7. If $P(A) = 4/5$ and $P(A \cap B) = 7/10$, then $P(B|A)$ is:

- (A) $1/10$ (B) $1/8$
(C) $7/8$ (D) $17/20$

8. If $A = [[0, a, -3], [2, 0, -1], [b, 1, 0]]$ is skew-symmetric, find a and b:

- (A) $a=-2, b=3$ (B) $a=2, b=3$
(C) $a=-2, b=-3$ (D) $a=2, b=-3$

9. $f(x) = [x]$ (Greatest Integer Function) is continuous at:

- (A) $x=1$ (B) $x=1.5$
(C) $x=-2$ (D) $x=4$

10. $\int_{-1}^1 (|x-2|)/(x-2) dx$ ($x \neq 2$) is:

- (A) 1 (B) -1
(C) 2 (D) -2

11. LPP objective function $Z = ax + by$ has maximum value:

- (A) Only origin (B) Only axes
(C) Corner points (D) Center

12. $|a|=3$, $|b|=2/3$ and $(a \times b)$ is a unit vector, then angle is:

- (A) $\pi/6$
(C) $\pi/3$

- (B) $\pi/4$
(D) $\pi/2$

13. General solution of $y \, dx - x \, dy = 0$ is:

- (A) $xy=C$
(C) $y=Cx$

- (B) $x=Cy$
(D) $y=Cx^2$

14. Rate of change of area of circle w.r.t radius r when $r=6\text{cm}$:

- (A) 10π
(C) 8π

- (B) 12π
(D) 11π

15. **A:** $R=\{(1,1),(2,2),(3,3),(1,2),(2,3)\}$ on $A=\{1,2,3\}$ is reflexive. **R:** $(a,a) \in R$ for all $a \in A$.

- (A) A,R True, Expl
(C) A True, R False

- (B) A,R True, No Expl
(D) A False, R True

16. **A:** $\int_{-a}^a f(x)dx=0$ if $f(x)$ is odd. **R:** $f(x)=\sin^7 x$ is odd.

- (A) A,R True, Expl
(C) A True, R False

- (B) A,R True, No Expl
(D) A False, R True

SECTION B: VERY SHORT ANSWER (2 MARKS EACH)

17. Find domain and range (principal value branch) of $f(x) = \cos^{-1}(x^2 - 4)$.

18. Find k if $f(x) = k \cos x / (\pi - 2x)$ for $x \neq \pi/2$ and $f(\pi/2)=3$ is continuous at $x=\pi/2$.

19. Find unit vector perpendicular to both $(a+b)$ and $(a-b)$ where $a = i+j+k$, $b = i+2j+3k$.

20. A coin is tossed 3 times. Find the probability of getting at least two heads.

21. Evaluate: $\int [\sin x / \sin(x-a)] \, dx$ OR $\int \log(\sin x) \cdot \cot x \, dx$.

SECTION C: SHORT ANSWER (3 MARKS EACH)

22. Solve the differential equation: $(x^2 + y^2)dx - 2xy \, dy = 0$. (Homogeneous type)

23. Evaluate: $\int_0^{\pi/2} [\sqrt{\sin x} / (\sqrt{\sin x} + \sqrt{\cos x})] \, dx$.

24. Show that $R = \{(a, b) : a \leq b^2\}$ on Real Numbers is neither Reflexive nor Symmetric nor Transitive.

25. Find the coordinates of the **foot of the perpendicular** from $(0, 2, 3)$ to the line $(x+3)/5 = (y-1)/2 = (z+4)/3$.

26. Two cards drawn from 52 cards. Find probability distribution of the number of aces.

27. If $a = 2i+2j+3k$, $b = -i+2j+k$, $c = 3i+j$ and $(a+\lambda b)$ is perpendicular to c , find λ .

SECTION D: LONG ANSWER (5 MARKS EACH)

28. Solve using matrix method: $x - 2y = 10$; $2x - y - z = 8$; $-2y + z = 7$. **OR** Find A^{-1} for $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$ and solve context.

29. Find shortest distance between $r=(i+2j+k)+\lambda(i-j+k)$ and $r=(2i-j-k)+\mu(2i+j+2k)$.

30. Using integration, find the area of the region bounded by the ellipse $x^2/16 + y^2/9 = 1$. **OR** Area between $y^2=4x$ and $x=3$.

31. Solve graphically: Maximize $Z = 50x + 70y$ subject to $2x+y \leq 8$, $x+2y \leq 10$, $x, y \geq 0$.

SECTION E: CASE STUDIES (4 MARKS EACH)

32. **Probability (Bayes' Theorem):** Plane late probabilities based on turbulence: Severe(0.1), Moderate(0.5), Light(0.4). Late given condition: S(0.6), M(0.4), L(0.1).

(i) Find total probability plane is late. (ii) Given late, find prob. it was Moderate.

33. **Derivatives (Optimization):** Box made from 18cm tin square by cutting corner squares (x) and folding.

(i) Express Volume V in terms of x . (ii) Find dV/dx . (iii) Find x for maximum volume.

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